Image alignment & Sea ice drift

Dr. Michel Tsamados

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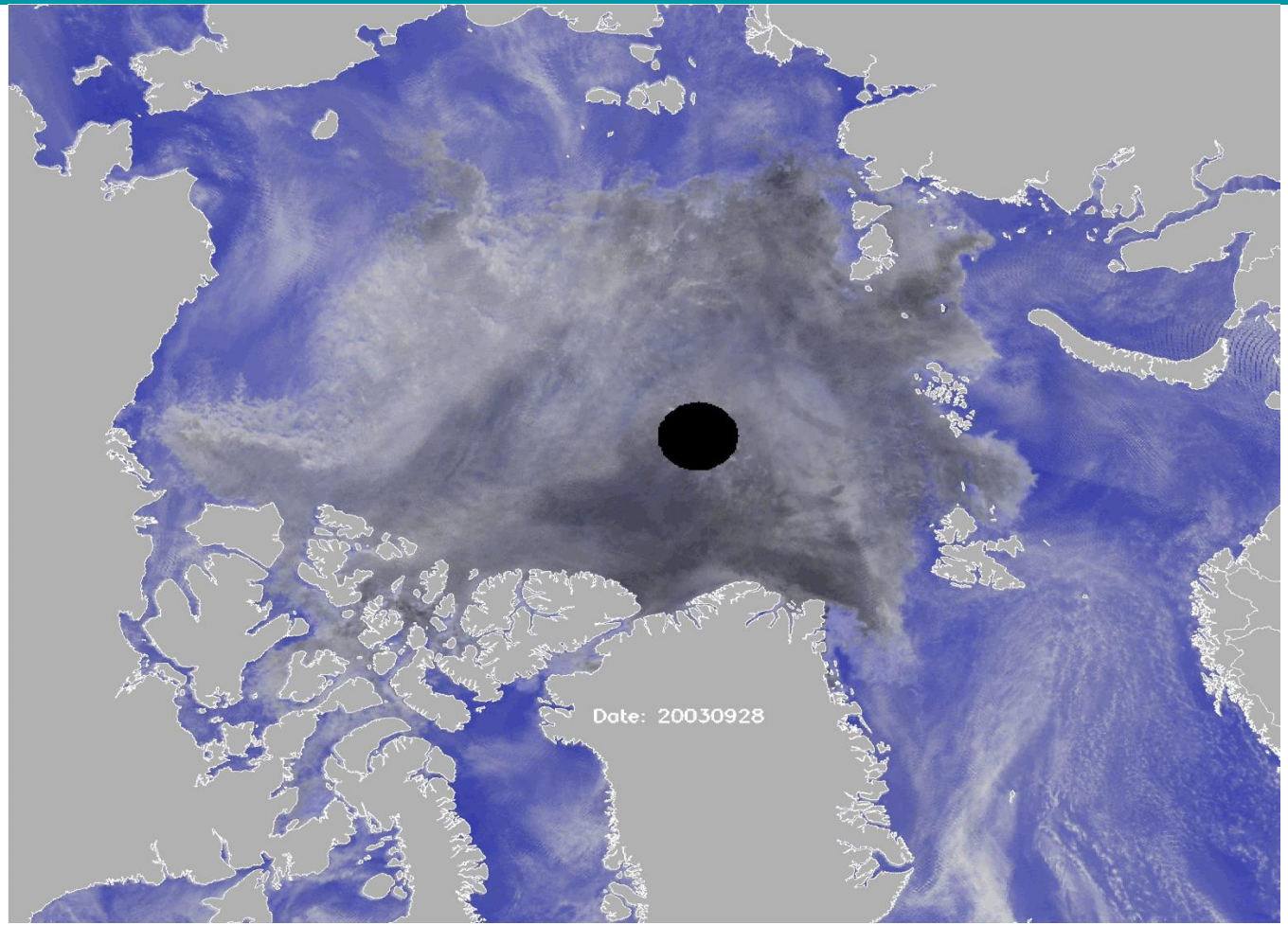
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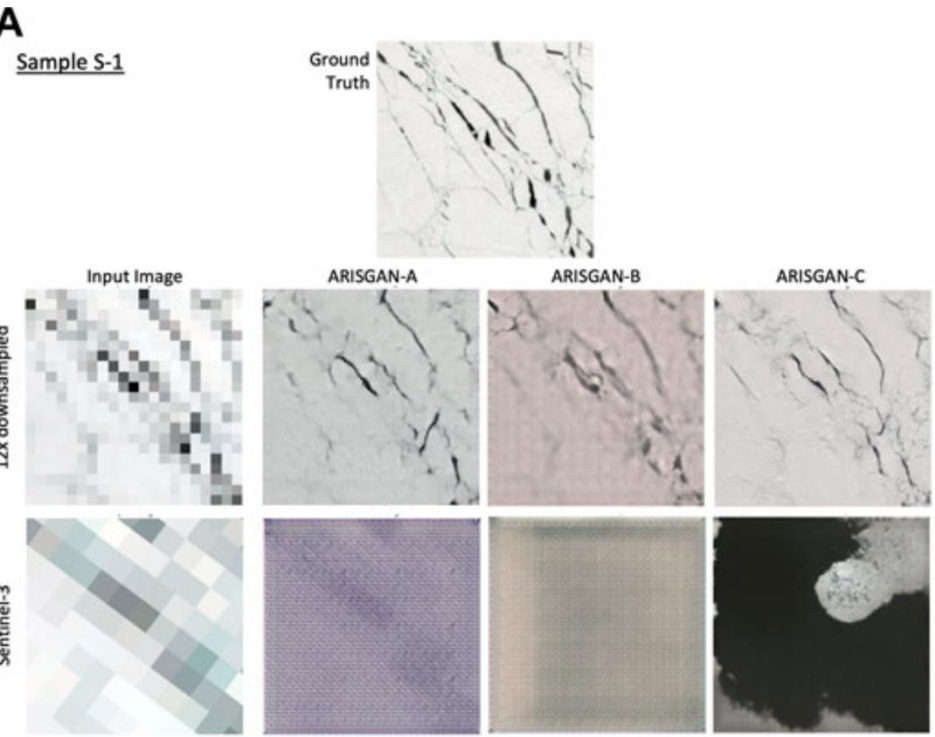
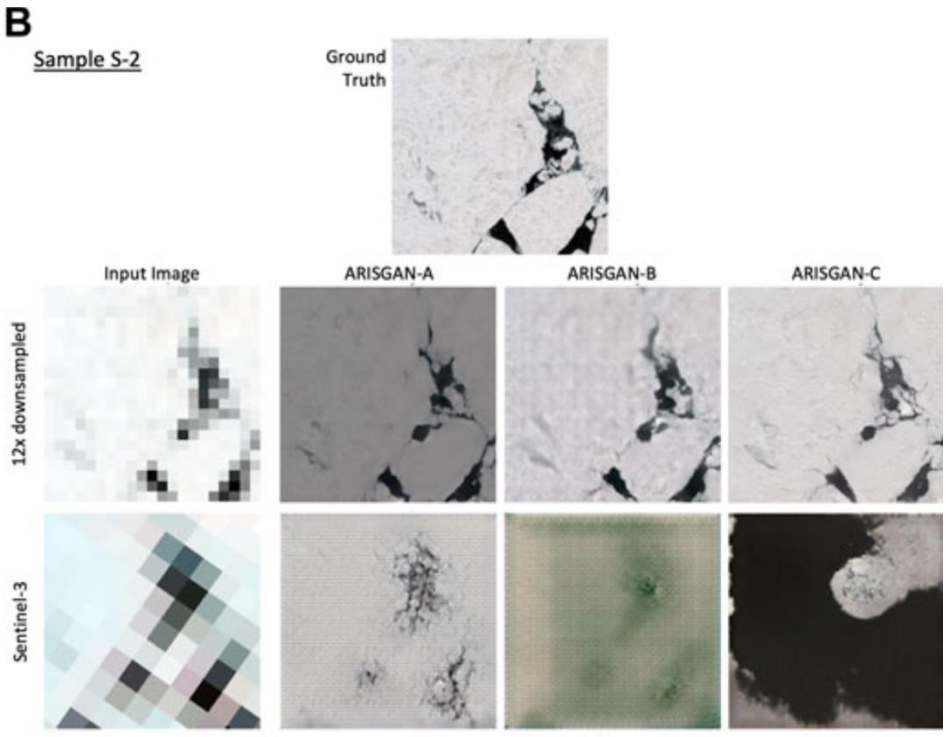
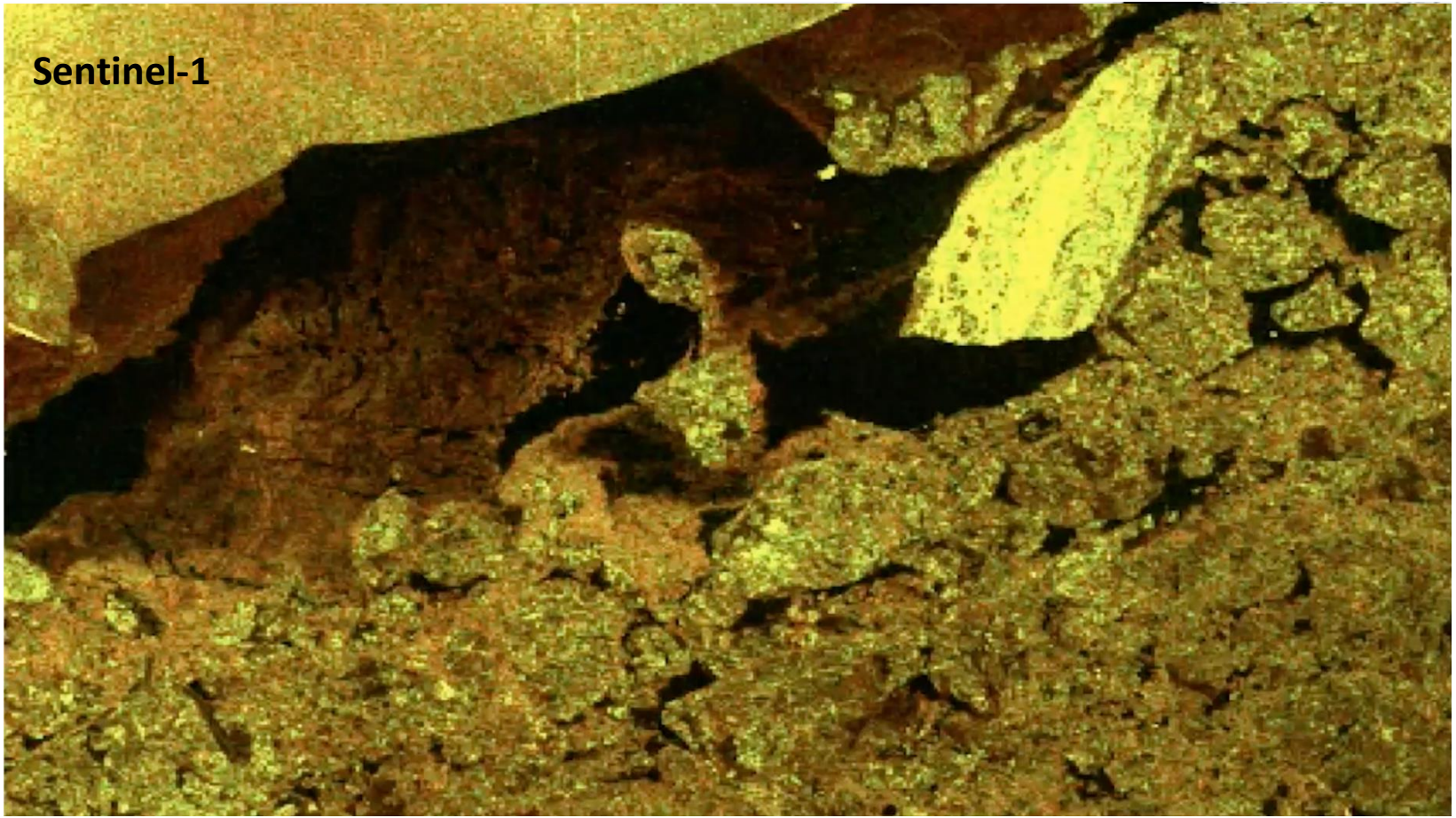

A
Sample S-1
Ground Truth
Input Image
ARISGAN-A
ARISGAN-B
ARISGAN-C
12x downsampled
Sentinel-3

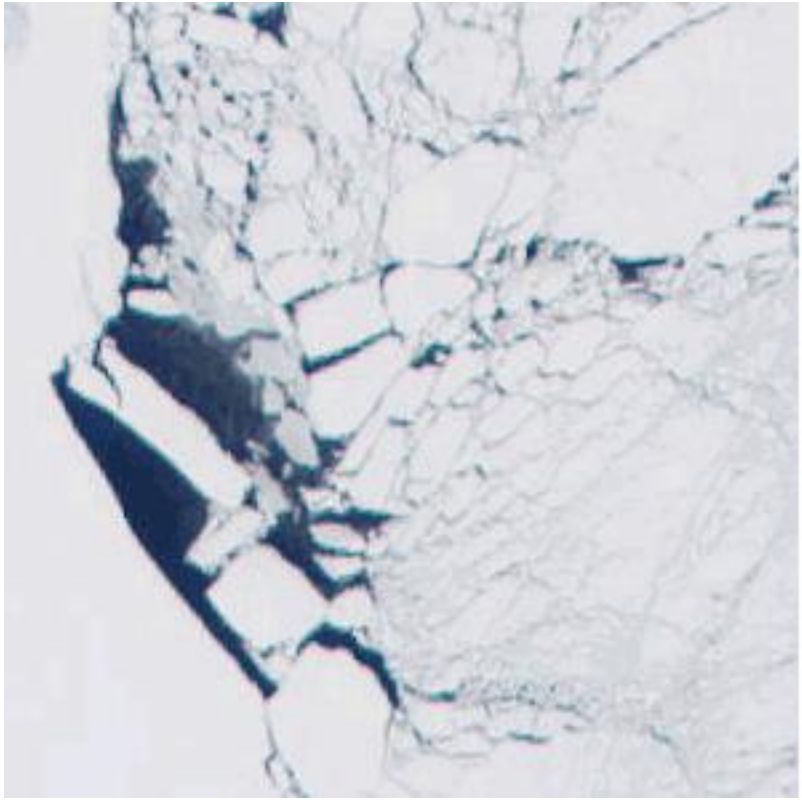

B
Sample S-2
Ground Truth
Input Image
ARISGAN-A
ARISGAN-B
ARISGAN-C
12x downsampled
Sentinel-3

Au et al, 2024

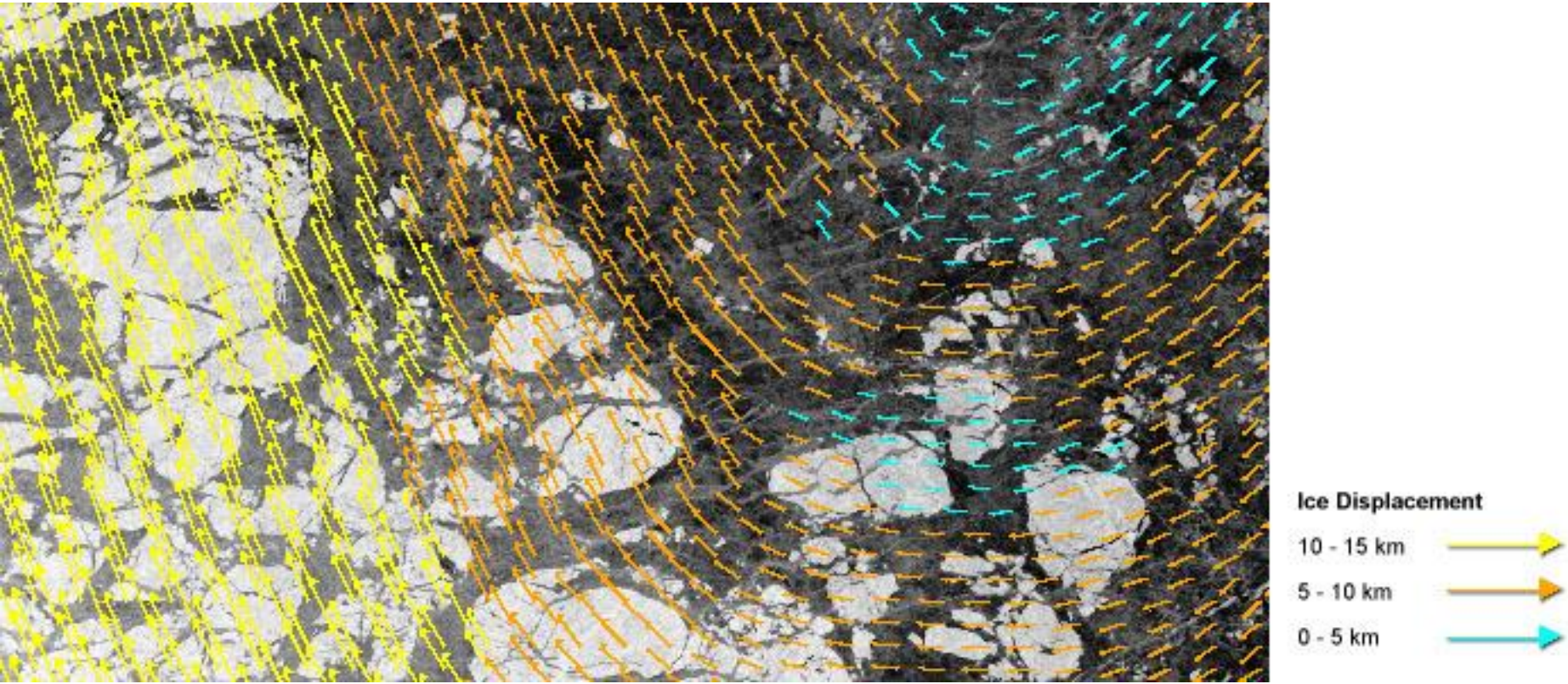
https://www.youtube.com/watch?v=KdaQgFGIxvI&t=1749s



https://www.youtube.com/watch?v=KdaQgFGIxvI&t=1749s



http://www.globice.info/Section.php?pid=27



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Files

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  - simple.ipynb
  - simple.py
- sea_ice_drift
  - __init__.py
  - ftlib.py
  - lib.py

sea_ice_drift / examples / drift_from_arrays.ipynb

akorosov add drift_from_arrays.ipynb 62efc5c · 3 years ago History

Preview | Code | Blame | 477 lines (477 loc) · 1.22 MB | Raw

In [1]:

```
import numpy as np
import matplotlib.pyplot as plt
import cartopy
import cartopy.crs as ccrs
import cartopy.feature as cfeature

from nansat import Nansat, Domain, NSR

from sea_ice_drift.lib import get_spatial_mean, get_uint8_image
from sea_ice_drift.ftlib import feature_tracking
from sea_ice_drift.pmlib import pattern_matching

%matplotlib inline
```

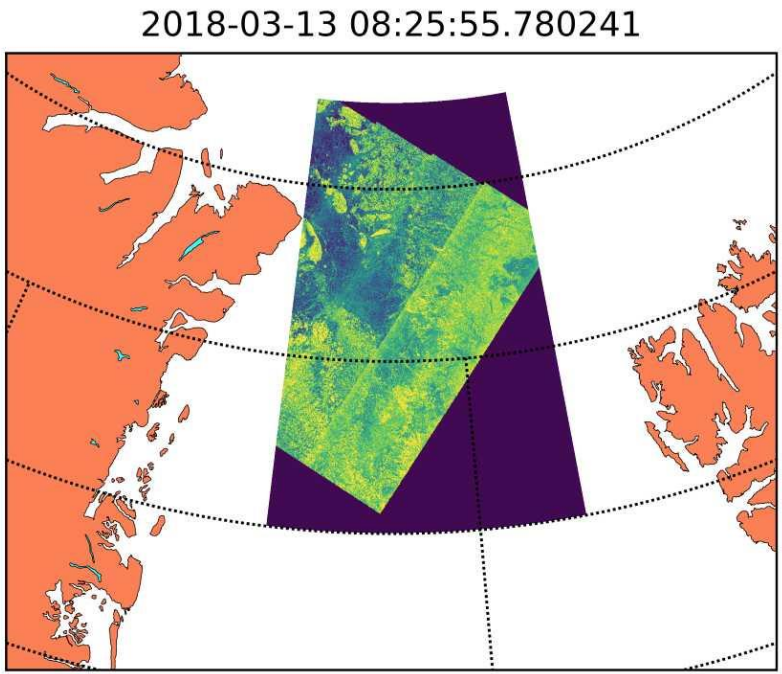
In [ ]:

```
# Download arrays
!wget https://github.com/nansencenter/sea_ice_drift_test_files/raw/master/S1B_EW_GRDM_1SDH_20200123T120
```

https://github.com/nansencenter/sea_ice_drift/blob/master/examples/drift_from_arrays.ipynb

Part 3: Image preprocessing: e.g. collocation and alignment using pattern matching and image tracking algorithms (Week 4 + Week 6)

Sea ice velocities from Sentinel1 pair of images (Fram)



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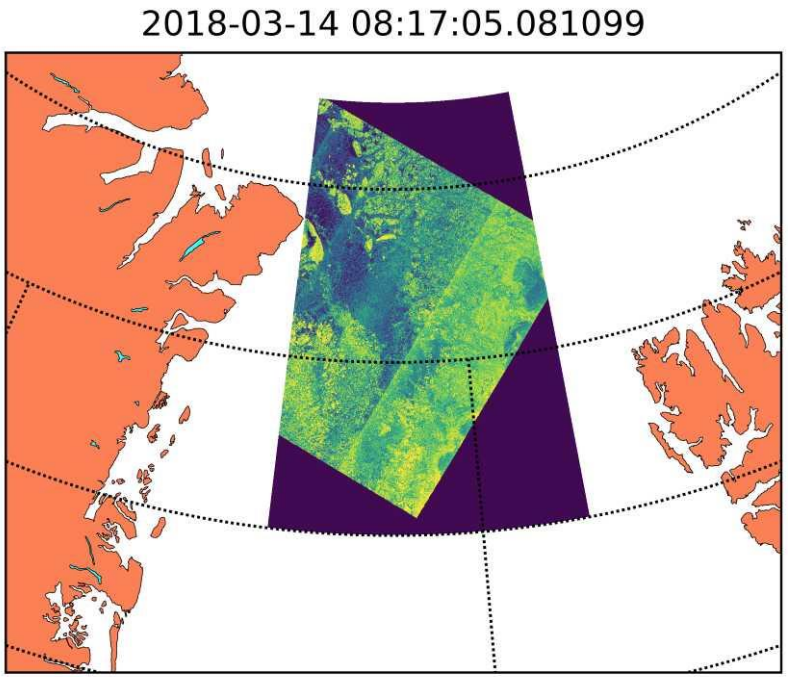
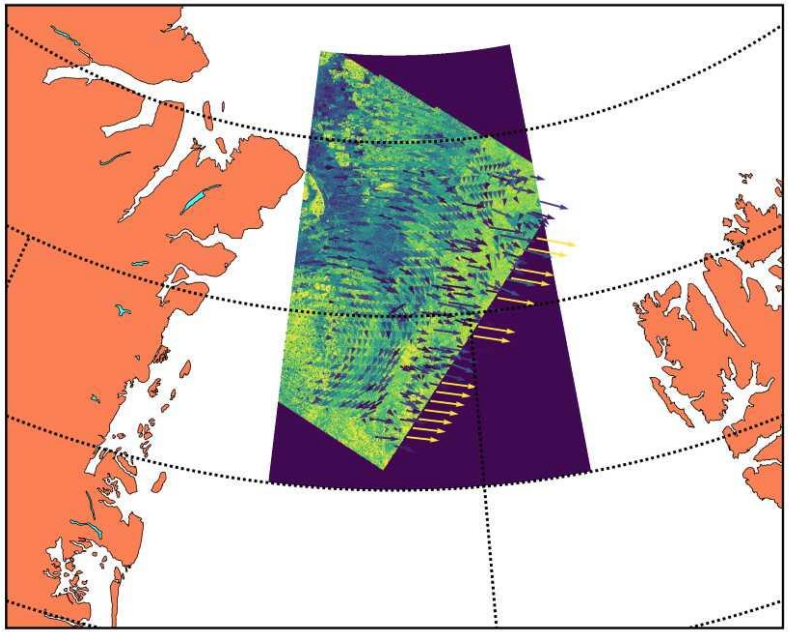


Part 3: Image preprocessing: e.g. collocation and alignment using pattern matching and image tracking algorithms (Week 4 + Week 6)

Sea ice velocities from Sentinel1 pair of images (Fram)

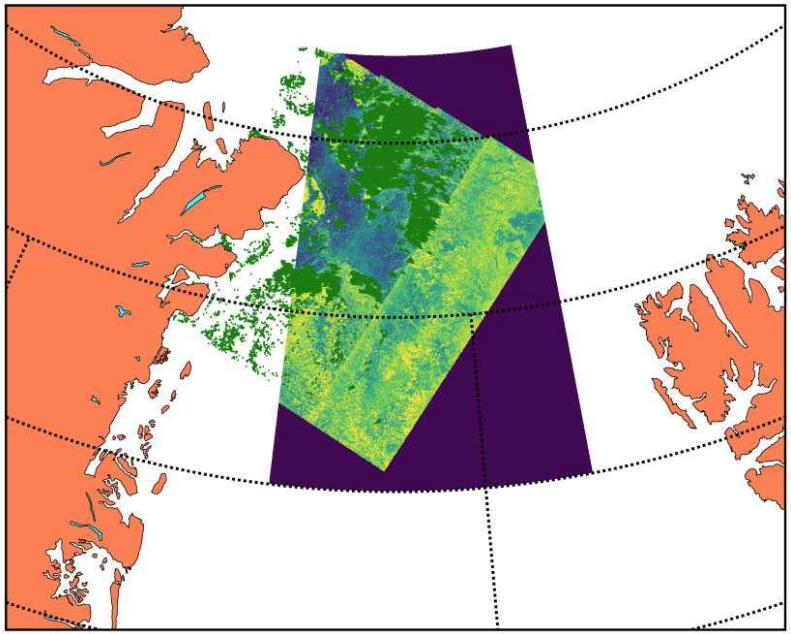
-> Pattern Matching

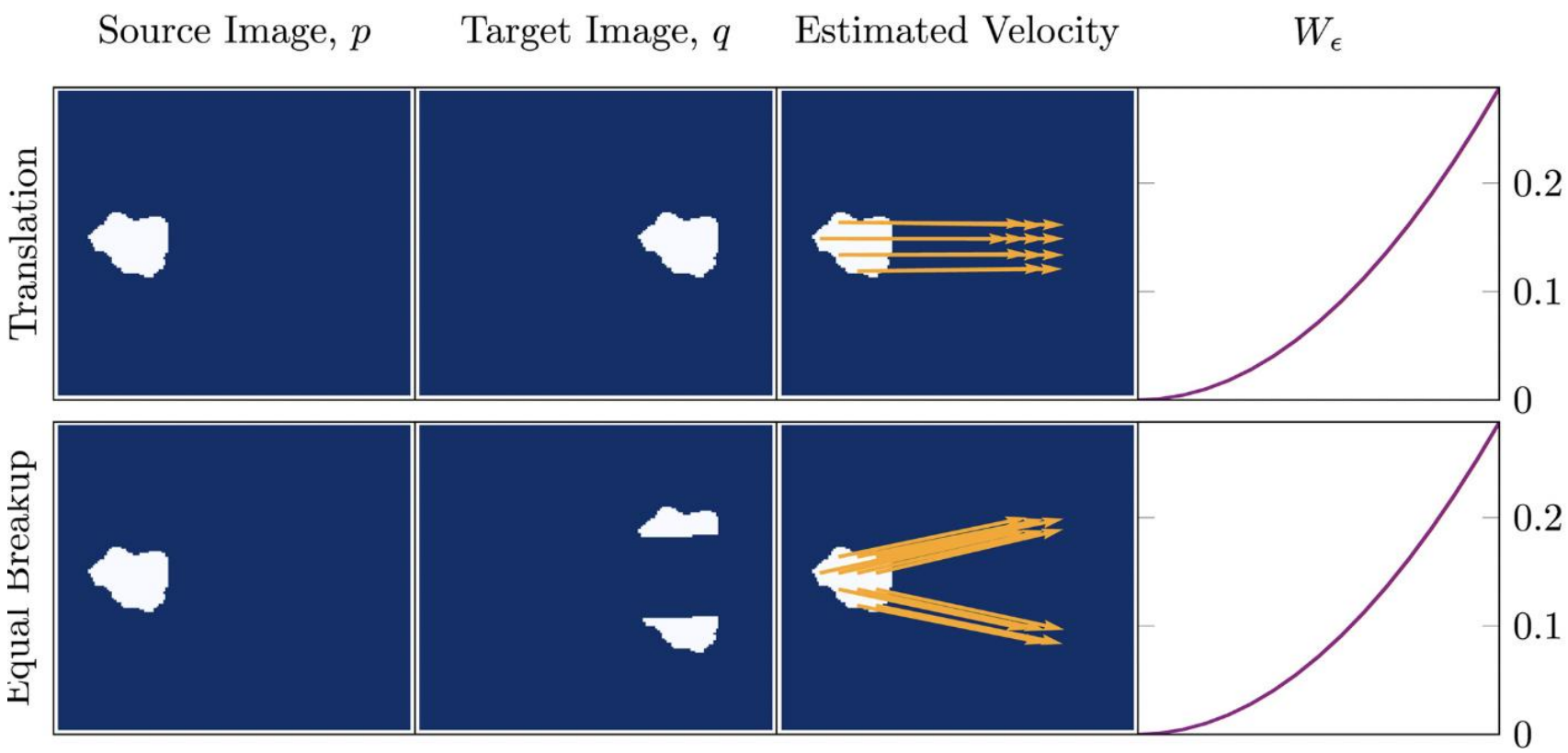


Part 3: Image preprocessing: e.g. collocation and alignment using pattern matching and image tracking algorithms (Week 4 + Week 6)

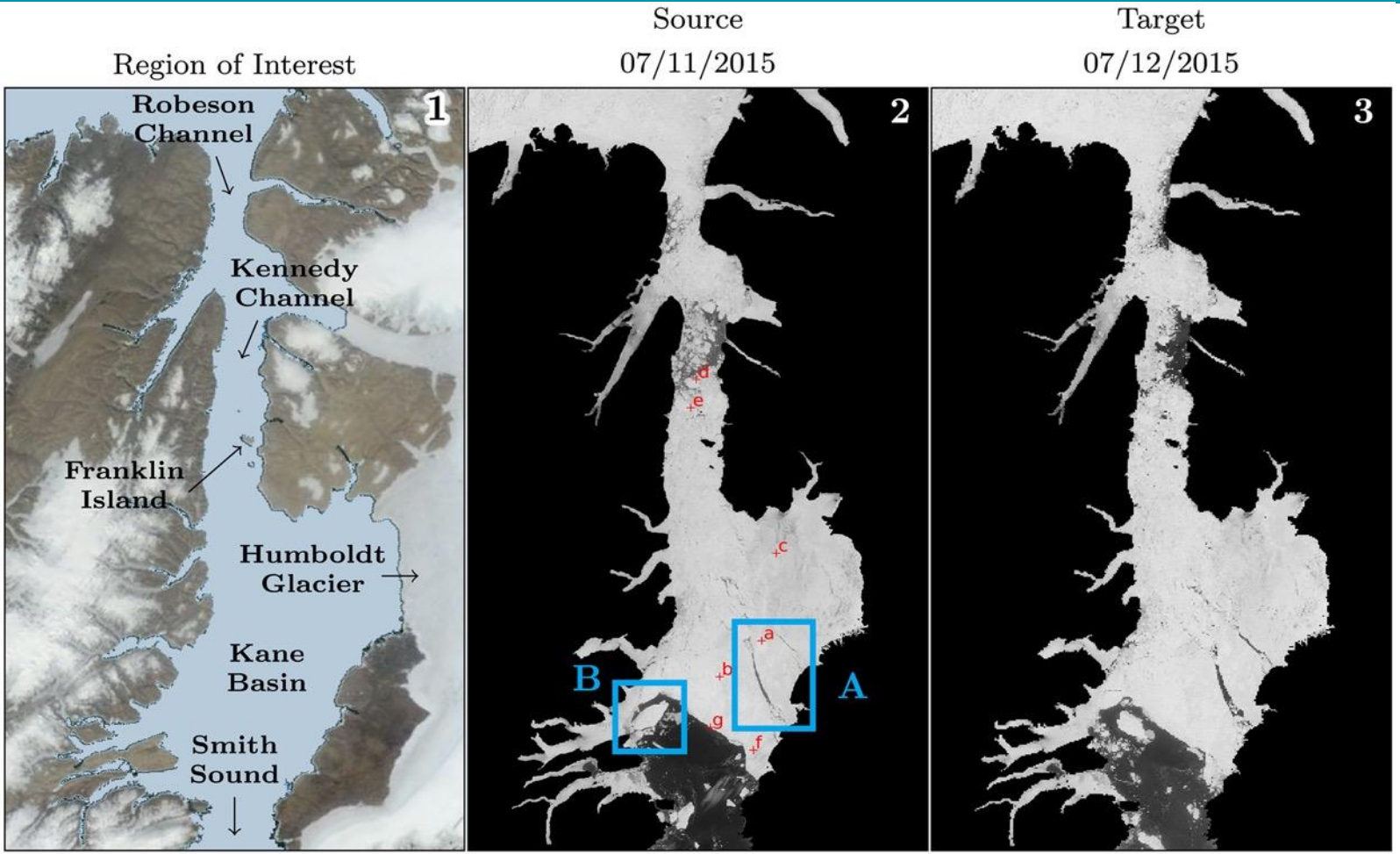
Sea ice velocities from Sentinel1 pair of images (Fram)

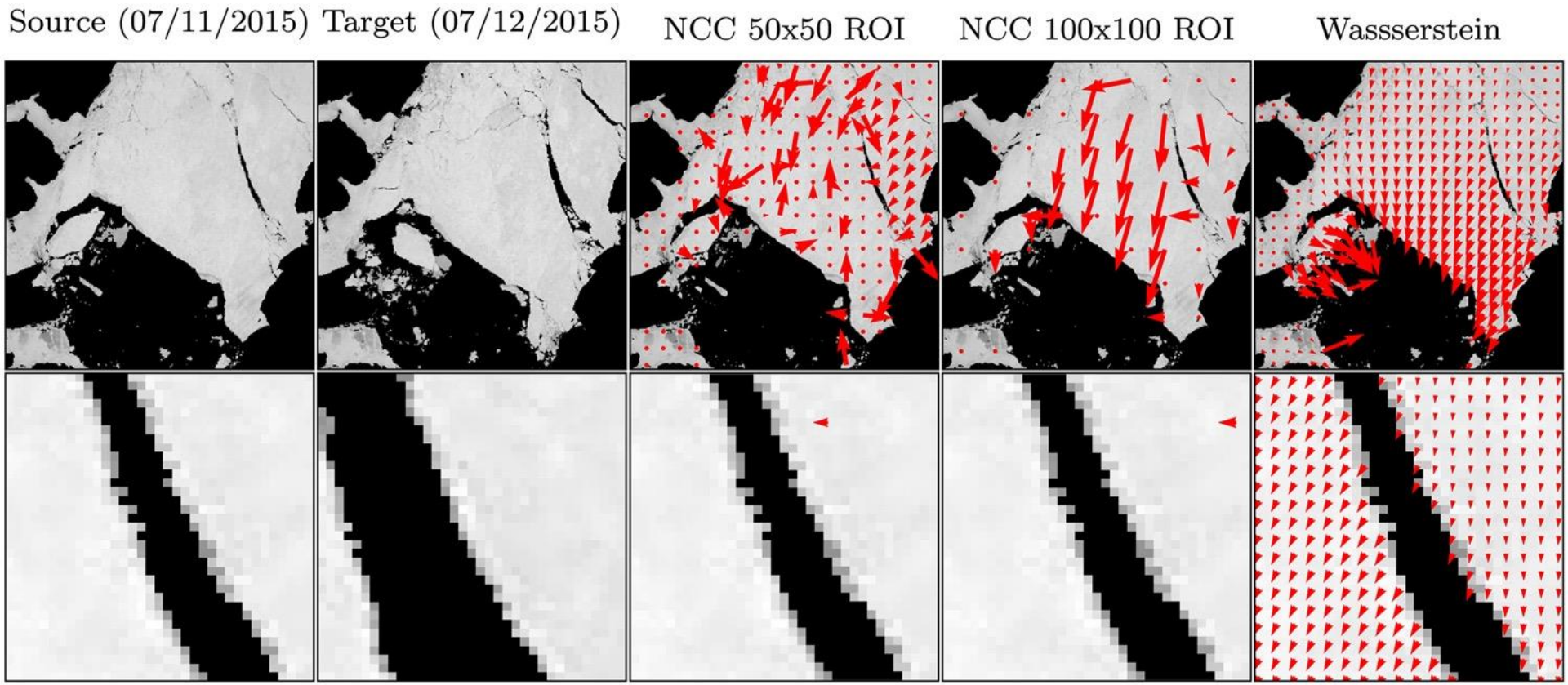
-> Feature tracking





https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/2019GL083037


Region of Interest
Source
07/11/2015
Target
07/12/2015
1
2
3
Robeson Channel
Kennedy Channel
Franklin Island
Humboldt Glacier
Kane Basin
Smith Sound
a
b
c
d
e
f
g
A
B


Source (07/11/2015)
Target (07/12/2015)
NCC 50x50 ROI
NCC 100x100 ROI
Wassserstein